

# Comparing Rates of Change

## Answer Key

Grade 8 / Pre-Algebra

### Quick Answers

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|---------------------------------------------------------|-------------------------------------------------------------|
| 1. (a) change in cost = 72, (b) dollars per month = 12  | son = <                                                     |
| 2. dollars per month = 9                                | 8. (a) change in pages left = -135, (b) pages per day = -15 |
| 3. rate of change of line P = 3                         | 9. (a) litres per minute = 5, (b) comparison = >            |
| 4. (a) metres per minute = 150, (b) comparison = <      | 10. metres per month = $\frac{3}{4}$                        |
| 5. percent per hour = -6                                | 11. (a) litres per minute = 12, (b) comparison = <, —       |
| 6. rate of change of line Q = -2                        | 12. (a) km per hour = 55, (b) comparison = <, —             |
| 7. (a) rate of change of line R = 2, (b) comparison = — |                                                             |
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### What is verified

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10 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 11, 12. The worked solution states what a correct response must contain.

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### Curriculum

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**Level:** Grade 8 / Pre-Algebra

8.EE.B.5 – problems 1, 4, 8, 9, 11, 12

8.EE.B.6 – problems 2, 3, 5, 6, 7, 10

*Difficulty 1–4 of 5 across 21 tagged checks.*

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- 1.** Gym table: months 0, 2, 4, 6 against total cost \$30, \$54, \$78, \$102. (a) Change in cost from month 0 to month 6. (b) Rate of change in dollars per month.

**Solution (a):** Read the two end rows: 102 dollars at month 6 against 30 dollars at month 0, so the change is  $102 - 30$ . 72

**Solution (b):** The change in cost goes on top and the change in months underneath:

$$\frac{72}{6 - 0}.$$
 $m = 12$ 

- 2.** Second gym:  $C = 9x + 45$  dollars for  $x$  months. Rate of change?

**Solution:** Increase  $x$  by one month. The cost goes from  $9x + 45$  to  $9(x + 1) + 45$ , and the difference is 9 dollars — the number multiplying  $x$ . The \$45 joining fee never changes, so it does not affect the rate. 9

- 3.** Line  $P$  through  $(0, 2)$  and  $(4, 14)$ . Rate of change?

**Solution:** From the first marked point to the second,  $y$  rises  $14 - 2 = 12$  while  $x$  runs  $4 - 0 = 4$ , so the rate is  $12/4$ .  $m = 3$

**4.** Ana: 0 minutes / 0 metres and 9 minutes / 1350 metres. Ben:  $d = 160t$ . (a) Ana's rate. (b) Compare.

**Solution (a):**  $\frac{1350 - 0}{9 - 0}$  metres per minute.  $m = 150$

**Solution (b):** Ben's equation has 160 multiplying  $t$ , so his rate is 160 metres per minute. Comparing Ana's rate with Ben's:  $150 < 160$

**5.** Battery:  $B = 45 - 6t$  percent after  $t$  hours. Rate of change?

**Solution:** Each extra hour subtracts another 6 percent, so the rate is negative: the charge falls by 6 percentage points per hour.  $-6$

**6.** Line  $Q$  through  $(0, 10)$  and  $(5, 0)$ . Rate of change?

**Solution:**  $y$  falls  $0 - 10 = -10$  while  $x$  runs  $5 - 0 = 5$ , so the rate is  $-10/5$ . The line falls from left to right, which agrees with the negative sign.  $m = -2$

**7.** Line  $R$  through  $(0, 1)$  and  $(6, 13)$ ; second line  $y = 3x$ . (a) Rate of  $R$ . (b) Compare.

**Solution (a):**  $\frac{13 - 1}{6 - 0} = \frac{12}{6}$ .  $m = 2$

**Solution (b):** The second line's equation has 3 multiplying  $x$ , so its rate is 3. Comparing the rate of  $R$  with it:  $2 < 3$

**8.** Reading log: days 0, 3, 6, 9 against pages left 200, 155, 110, 65. (a) Change in pages left. (b) Rate of change in pages per day.

**Solution (a):**  $65 - 200$  pages.  $-135$

**Solution (b):**  $\frac{-135}{9 - 0}$  pages per day; the book is getting shorter, so the rate is negative.  $m = -15$

**9.** Machine A makes 3 items per minute (line  $P$ ). Machine B: 0 minutes / 5 items and 8 minutes / 45 items. (a) B's rate. (b) Compare.

**Solution (a):**  $\frac{45 - 5}{8 - 0}$  items per minute.  $m = 5$

**Solution (b):** Comparing machine B's rate with machine A's rate of 3:  $5 > 3$

**10.** Sapling:  $y = \frac{3}{4}x + 2$  metres after  $x$  months. Rate of change?

**Solution:** The number multiplying  $x$  is  $\frac{3}{4}$ , so each extra month adds three quarters of a metre. The starting height of 2 metres does not change the rate.  $m = \frac{3}{4}$

**11.** Pool 1: 0 minutes / 20 litres and 6 minutes / 92 litres. Pool 2:  $L = 14t + 20$ . (a) Pool 1's rate. (b) Compare. (c) Explain from a graph.

**Solution (a):**  $\frac{92 - 20}{6 - 0}$  litres per minute.  $m = 12$

**Solution (b):** Pool 2's equation has 14 multiplying  $t$ , so its rate is 14 litres per minute. Comparing pool 1 with pool 2:  $12 < 14$

**Solution (c) — instructor-judged.** A correct explanation says the steeper line is the faster pool, and connects steepness to the rate: pool 2 climbs 14 litres for every minute

across while pool 1 climbs only 12, so pool 2's line is steeper. Both pools start at the same 20 litres, so pool 2's line is above pool 1's from the first minute onward. "Look at the graph" with no statement of what to look at does not earn credit.

**12.** Service X: 0 hours / 0 km and 5 hours / 275 km. Service Y:  $d = 60t$ . (a) X's rate. (b) Compare. (c) Explain from a graph.

**Solution (a):**  $\frac{275 - 0}{5 - 0}$  kilometres per hour.

$$m = 55$$

**Solution (b):** Service Y's equation has 60 multiplying  $t$ , so its rate is 60 km per hour. Comparing X with Y:

$$55 < 60$$

**Solution (c) — instructor-judged.** A correct explanation says both services start at zero kilometres, so their graphs share the same starting point and the faster one is simply whichever line is steeper — equivalently, whichever line is higher at any chosen time. Reading both heights at one common hour and comparing them earns full credit. Restating "Y is faster" with no graphical reason does not.